HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and scaled **automated moderation systems** and **abuse detection capabilities** for AI agents, processing over **5,000 daily events**.
- Partnered with the Investigations team to analyze emerging abuse patterns and iteratively enhanced **detection mechanisms**, reducing false positives by **18%**.
- Owned the architecture, deployment, and reliability of a growing ecosystem of **internal safety services**, ensuring **99.5% availability**.
- Collaborated with ML engineers to integrate and operate **fine-tuned ML models** within production safety systems, managing inference endpoints at scale.
- Developed **backend services and REST APIs** using Python and Flask to support moderation workflows and policy enforcement across product platforms.
- Translated product, legal, and safety policy requirements into scalable engineering solutions, working closely with **non-technical stakeholders**.
- Containerized internal safety services using **Docker** and deployed them to a **Kubernetes** cluster on AWS, managing CI/CD pipelines with Jenkins.

SKILLS

Languages & Frameworks – Python, Flask, FastAPI, Java, Spring Boot, React, Angular, TypeScript, SQL, R

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Kubernetes, Jenkins, Git, CI/CD

Databases – PostgreSQL, MySQL, Oracle, Redis

AI & ML – ML model integration, LLM guardrails, Abuse detection, Moderation systems, Prompt Engineering

Observability & Monitoring – Splunk, Dynatrace, Prometheus, Grafana

Data Analytics & BI – Power BI, Tableau, DAX, Power Query, pandas, NumPy, scikit-learn

PROJECTS

AI Agent Safety Guardrails Implementation

- Developed Python-based backend services to implement **safety guardrails** for AI-generated content, filtering for policy violations and unsafe language.
- Integrated with a proprietary **LLM API** and implemented real-time content analysis, achieving a detection rate of **92%** for harmful outputs.
- Built a **FastAPI** inference service that processed over **1,500 requests per minute**, providing moderation feedback to downstream content generation systems.

Production Anomaly Detection System

- Engineered a Python data pipeline using **pandas** and **NumPy** to process logs from enterprise applications, identifying anomalous patterns.
- Trained a **scikit-learn** anomaly detection model on a dataset of **~150,000 log entries**, achieving a **0.82 F1-score** for outlier identification.
- Deployed the anomaly detection model as a **Flask** service, enabling real-time alerting for critical system events and reducing manual review effort by **30%**.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)